

## Exercise Sheet 2: Multiple Quantum Systems

Use the ordered basis  $|00\rangle, |01\rangle, |10\rangle, |11\rangle$  for two qubits, and write  $|\pm\rangle = (|0\rangle \pm |1\rangle)/\sqrt{2}$ . In Questions 1 and 2, select all correct answers and briefly justify them.

**Question 1.** Consider the two-qubit state

$$\rho_{AB} = \frac{1}{2}(|00\rangle\langle 00| + |11\rangle\langle 11|).$$

Which of the following describe  $\rho_{AB}$ ?

- (A) Pure.
- (B) Separable.
- (C) Mixed.
- (D) Entangled.

**Question 2.** Consider the two-qubit state

$$\rho_{AB} = \frac{1}{2}(|00\rangle + |11\rangle)(\langle 00| + \langle 11|).$$

Which of the following describe  $\rho_{AB}$ ?

- (A) Pure.
- (B) Entangled.
- (C) Mixed.
- (D) A purification of  $\frac{1}{2}I_A$ .

**Question 3.** Let  $\tau_{AB} = |\psi\rangle\langle\psi|$ , where

$$|\psi\rangle = \frac{1}{\sqrt{3}}(|01\rangle - |00\rangle + |11\rangle).$$

Which operator is  $\tau_A = \text{tr}_B[\tau_{AB}]$ ?

$$\begin{aligned} A &= |1\rangle\langle 1|, & B &= \frac{1}{3}(2|0\rangle\langle 0| + |1\rangle\langle 1| + |0\rangle\langle 1| + |1\rangle\langle 0|), \\ C &= |1\rangle\langle 1| + |0\rangle\langle 1| + |1\rangle\langle 0|, & D &= \frac{1}{3}(|0\rangle\langle 0| + 2|1\rangle\langle 1| - |0\rangle\langle 1| - |1\rangle\langle 0|). \end{aligned}$$

**Question 4.** Let

$$|u\rangle = \frac{|0\rangle + i|1\rangle}{\sqrt{2}}, \quad |v\rangle = \frac{\sqrt{3}}{2}|0\rangle + \frac{1}{2}|1\rangle, \quad X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

- (a) Expand  $|u\rangle \otimes |v\rangle$  in the computational basis and write its column vector.
- (b) Write the matrices  $X \otimes I_2$  and  $I_2 \otimes X$ . Compute their action on  $|u\rangle \otimes |v\rangle$ , and identify which qubit each operator changes.

**Question 5.** Consider the pure two-qubit state

$$|\chi\rangle = \frac{1}{\sqrt{8}}(\sqrt{3}|00\rangle + \sqrt{3}|01\rangle + |10\rangle - |11\rangle).$$

- (a) Find a Schmidt decomposition and the Schmidt rank of  $|\chi\rangle$ .
- (b) Compute both reduced states and their eigenvalues.
- (c) Is  $|\chi\rangle$  a product state, entangled, or maximally entangled? Explain using the Schmidt coefficients.

**Question 6.** Let  $A$  be a three-dimensional system with state

$$\rho_A = \begin{pmatrix} 2/3 & 0 & 0 \\ 0 & 1/3 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

- (a) Construct a purification  $|\phi_{AR}\rangle$  using a qubit reference system  $R$ , and verify that tracing out  $R$  gives  $\rho_A$ .
- (b) Replace the reference basis vectors  $|0\rangle, |1\rangle$  in your construction by  $|+\rangle, |-\rangle$ . Verify that this gives a different purification of the same state.
- (c) What is the smallest possible dimension of  $R$ ? Explain why the zero eigenvalue of  $\rho_A$  does not require another reference basis vector.

**Question 7.** Alice and Bob share  $|\Phi^+\rangle = (|00\rangle + |11\rangle)/\sqrt{2}$ . Alice measures in the  $X$  basis  $\{|+\rangle, |-\rangle\}$ . Bob chooses either the  $Z$  basis  $\{|0\rangle, |1\rangle\}$  or the  $X$  basis.

- (a) Find the joint outcome distribution for each of Bob's two choices.
- (b) Find Alice's marginal distribution in both cases.
- (c) Compute Alice's conditional probability of  $+$  given that Bob obtains 0 in the  $Z$  basis, and given that he obtains  $+$  in the  $X$  basis. Explain why Bob cannot communicate his choice of basis through these correlations alone.

**Question 8.** Consider the four Bell vectors

$$|\Phi^\pm\rangle = \frac{|00\rangle \pm |11\rangle}{\sqrt{2}}, \quad |\Psi^\pm\rangle = \frac{|01\rangle \pm |10\rangle}{\sqrt{2}}.$$

- (a) Check that they form an orthonormal basis of the two-qubit space.
- (b) Compute their reduced states. Can Alice distinguish which Bell state was prepared by measuring her qubit alone?
- (c) Starting from  $|\Phi^+\rangle$ , apply  $I_2 \otimes X$ ,  $I_2 \otimes Z$ , and  $I_2 \otimes (XZ)$ , where  $Z = \text{diag}(1, -1)$ . Which Bell vector results in each case?